

A Physical-Layer Intrusion Detection Framework for ADS-B Using Differential Constellation Trace

Yiğit Emir Işıklı

Avionic Cybersecurity Research Lab

ASELSAN

Ankara, Türkiye

yemirisikci@aselsan.com

Selami Utku Yücel

Avionic Cybersecurity Research Lab

ASELSAN

Ankara, Türkiye

selamiutkuy@aselsan.com

Mustafa Evcil

Avionic Cybersecurity Research Lab

ASELSAN

Ankara, Türkiye

mevcil@aselsan.com

Mustafa Emir Peker

Avionic Cybersecurity Research Lab

ASELSAN

Ankara, Türkiye

memirpeker@aselsan.com

İsa Can Babir

Avionic Cybersecurity Research Lab

ASELSAN

Ankara, Türkiye

isacanbabir@aselsan.com

Yiğit Çevik

Avionic Cybersecurity Research Lab

ASELSAN

Ankara, Türkiye

yigitcevik@aselsan.com

Abstract—Automatic Dependent Surveillance–Broadcast (ADS-B) plays a critical role in modern air traffic surveillance by enabling aircraft to periodically broadcast state information such as position, velocity, altitude, and identification. However, ADS-B was designed as an open broadcast protocol without authentication or encryption mechanisms, which makes it vulnerable to spoofing attacks, ranging from simple signal replay to sophisticated transmitter impersonation. Securing the widespread ADS-B infrastructure through protocol modifications or complex hardware updates is cost-prohibitive and highly impractical due to rigorous aviation certification processes. While attack scenarios can be simulated in isolated environments, executing cyberattacks under real-world conditions poses severe safety risks and is highly restricted by aviation regulations. To address these limitations, we propose an unsupervised physical-layer intrusion detection framework for ADS-B based on Differential Constellation Trace (DCT) features and convolutional autoencoders, trained only on benign transmissions. The proposed approach utilizes transmitter-specific signal characteristics introduced by hardware imperfections, while differential processing provides robustness to channel variations and reduces receiver-dependent effects, making it suitable for pulse position modulated ADS-B signals. The proposed method is developed using a real-world dataset comprising raw In-Phase and Quadrature (I/Q) traces from 160 distinct civil aircraft collected over a three-day period. Performance is evaluated against identity impersonation attacks derived from benign transmissions and modeled with realistic hardware imperfections, including Carrier Frequency Offset (CFO), I/Q imbalance, and additive white Gaussian noise (AWGN). Experimental results indicate a high detection performance with an F1-score exceeding 97%, demonstrating that the proposed method effectively identifies anomalous ADS-B transmissions without protocol modifications or reliance on prior knowledge of attack patterns.

Index Terms—ADS-B, DCT, radio frequency fingerprinting, avionics security, intrusion detection system

I. INTRODUCTION

Automatic Dependent Surveillance–Broadcast (ADS-B) is a cornerstone of modern air traffic surveillance and a key

enabler of next-generation operations, complementing conventional radar-based systems. It enables aircraft to periodically transmit status data, including position, velocity, altitude, and identification information [1]. Despite its central role in today's surveillance infrastructure, the protocol was specified without built-in cryptographic authentication or encryption mechanisms, allowing adversaries to generate messages that remain compliant with the ADS-B format [2]. In addition, the widespread availability of low-cost Software-Defined Radios (SDRs) further lowers the barrier to transmitting spoofed signals, raising concerns about operational safety.

In operational aviation, ADS-B is a broadcast-based surveillance technology governed by international regulations. As a result, unauthorized generation, modification, or manipulation of ADS-B signals is prohibited outside controlled and authorized test environments. This constraint makes it impractical to collect malicious transmissions from operational airspace, and therefore the evaluation of attack detection mechanisms cannot rely on real attack traffic. The dataset used in this study was constructed using open-source data from real operational flights for benign samples, while malicious signals were synthetically generated.

To address these limitations, physical-layer security methods can improve the resilience of ADS-B without requiring protocol modifications or relying on the decoded message contents. In this framework, Radio Frequency Fingerprinting (RFF) utilizes distinctive signal features that arise from the intrinsic hardware characteristics of individual transmitters. Variations introduced by manufacturing tolerances and component-level differences, such as oscillator stability, clock behavior, digital-to-analog conversion, and power amplification, produce minor yet repeatable imperfections in the emitted signal. These imperfections form device-specific signatures that can be observed at the receiver [3]. Because such characteristics are largely independent of the transmitted data, RFF is a natural fit for ADS-B scenarios where messages are openly

broadcast and cryptographic safeguards are absent.

This study proposes an unsupervised physical-layer intrusion detection framework for ADS-B based on Differential Constellation Trace (DCT) features and convolutional autoencoders. We select DCT features to reduce sensitivity to channel-induced variability and receiver-side impairments. Unlike amplitude or absolute phase-based descriptors, DCT captures relative transitions between consecutive constellation points, which remain more stable under gain fluctuations and constant phase offsets [4]. This property is well suited to ADS-B signals, where Pulse Position Modulation (PPM) produces sparse structures and varying amplitude patterns, while still enabling a consistent representation of transmitter-specific behavior for unsupervised intrusion detection. Although differential constellation-based fingerprinting has been explored in other wireless standards, such as IEEE 802.11 and Bluetooth [5], [6], its adaptation to the PPM signaling structure of ADS-B has received limited attention.

The performance of the proposed approach is assessed using a large-scale dataset gathered during a continuous three-day passive flight monitoring campaign [7]. The dataset contains raw In-Phase and Quadrature (I/Q) samples collected from 160 different aircraft, each identified by its International Civil Aviation Organization (ICAO) 24-bit address. The recorded packets are first aligned in time by applying cross-correlation with an ideal preamble template. After alignment, a fixed temporal offset is introduced to extract the portion of the signal that carries the ICAO field. This procedure ensures consistent segment selection across packets and limits distortions caused by timing variations in the subsequent feature extraction stage.

The main contributions of this work are as follows: a novel application of DCT features is introduced, specifically tailored for PPM structure of ADS-B to extract robust hardware-specific signatures; an unsupervised deep convolutional autoencoder framework is proposed, which eliminates the need for labeled attack data and enables the detection of zero-day spoofing vectors; a comparative analysis is provided, demonstrating that the proposed unsupervised approach outperforms baseline anomaly detection models in terms of reliability; the system is validated using a large-scale real-world dataset from 160 aircraft, augmented with realistic hardware impairments such as Carrier Frequency Offset (CFO), I/Q imbalance, and Additive White Gaussian Noise (AWGN). Experimental evaluations under realistic channel conditions indicate that the framework maintains high detection performance showcasing its reliability for operational aviation security without altering the existing protocol.

II. RELATED WORK

The Automatic Dependent Surveillance–Broadcast (ADS-B) system was designed as an open broadcast protocol without cryptographic authentication or encryption, which makes it vulnerable to spoofing, replay, and transmitter impersonation attacks [8]. The availability of low-cost Software-Defined Radios has further lowered the barrier to generating protocol-compliant but deceptive transmissions, increasing concern

for operational safety in civil aviation. Existing security approaches for ADS-B can be broadly categorized into cryptographic mechanisms, location-based verification, data-driven anomaly detection, and physical-layer radio frequency fingerprinting.

A. Cryptographic Mechanisms

Cryptographic protection schemes for ADS-B, including certificate-based authentication and public-key or symmetric-key methods, have been proposed to provide message integrity and source authentication [9], [10]. However, their deployment remains challenging due to the need for protocol modification, global standardization, key management at scale, and compatibility with existing avionics systems. As a result, cryptographic solutions have not been widely adopted in operational ADS-B environments.

B. Location-Based Approaches

Location-based verification techniques exploit physical signal properties to validate reported aircraft positions. Multilateration estimates transmitter location using time-difference-of-arrival measurements from multiple receivers, while Doppler-based approaches verify velocity consistency using frequency shift observations [11], [12]. These methods can detect spatially inconsistent transmissions but rely on dense ground infrastructure and favorable propagation conditions, and they are less effective against attacks originating from physically plausible locations or emulating legitimate flight behavior.

C. Data-Driven Anomaly Detection

Machine learning has been widely explored for detecting anomalous ADS-B behavior without protocol modification. Early work by Habler and Shabtai applied an LSTM-based encoder–decoder to flight trajectories, learning normal route patterns from legitimate data and identifying anomalies through reconstruction error [13]. Their subsequent work extended this approach to sequential airspace states, further improving detection reliability [14]. Wang et al. [15] proposed an LSTM-based spoofing detector operating on decoded message fields, while Ying et al. [16] employed a deep neural network on kinematic features. Chevrot [17] introduced a contextual autoencoder for multivariate ADS-B time-series, demonstrating the applicability of reconstruction-based deep learning to trajectory anomalies. Çevik and Akleyek [18] provided a comprehensive evaluation across eight distinct attack scenarios including spoofing, message injection, and virtual trajectory modification testing a broad set of classical and deep learning classifiers on kinematic features derived from the OpenSky Network, with tree-based ensemble methods emerging as the strongest performers.

Despite their effectiveness, trajectory- and kinematics-based detectors inherently assume that malicious activity manifests as deviations in flight behavior. Consequently, these approaches cannot reliably detect attacks in which adversarial transmissions remain protocol-compliant and kinematically

plausible, or where unauthorized transmitters impersonate legitimate aircraft. Such scenarios motivate security mechanisms that operate at the physical layer.

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D. Physical-Layer Security and RF Fingerprinting

Physical-layer security techniques aim to authenticate transmitters by exploiting device-specific signal characteristics arising from hardware imperfections, an approach commonly referred to as RFF or specific emitter identification [19]. These characteristics persist even when transmitted message content is identical, making RFF well suited to ADS-B, where data fields are publicly accessible. Several supervised RFF approaches have been proposed for ADS-B using raw I/Q samples or derived signal representations in combination with convolutional neural networks. While high classification accuracy has been reported under controlled conditions, these methods typically require labeled enrollment data for each aircraft and exhibit sensitivity to channel and receiver variations, limiting their applicability for intrusion detection in operational environments [20], [21]. Furthermore, ADS-B employs pulse position modulation with short, sparse bursts, which differs substantially from the continuous modulation schemes commonly assumed in conventional RFF techniques.

E. Summary and Research Gap

Existing ADS-B security mechanisms face complementary limitations. Cryptographic and location-based approaches are difficult to deploy at scale or are limited to detecting spatial inconsistencies. Trajectory-based anomaly detection methods cannot authenticate transmitters when adversarial signals remain behaviorally plausible. Most RF fingerprinting techniques rely on supervised learning and labeled enrollment data, which is impractical in ADS-B environments where malicious transmissions cannot be collected from live operational airspace and unknown attackers cannot be pre-registered. Additionally, raw I/Q-based features are sensitive to channel and receiver variability. To address these limitations, this work proposes an unsupervised physical-layer intrusion detection framework for ADS-B based on differential constellation trace features and convolutional autoencoders. The proposed approach learns legitimate transmitter behavior from unlabeled data, provides robustness to channel variations, and operates without protocol modification or additional infrastructure, complementing existing trajectory-based security mechanisms.

III. BACKGROUND

A. ADS-B

Automatic Dependent Surveillance–Broadcast operates in the 1090 MHz band using the 1090 Extended Squitter

(1090ES) format for most commercial and civil aviation applications [1]. The aircraft equipped with ADS-B Out periodically broadcast state information including position, velocity, altitude, and identification. These broadcasts are received by ground stations and nearby aircraft through the ADS-B In functionality. At the physical layer, ADS-B employs PPM with a data rate of 1 Mbps. Each transmission consists of an 8 μ s preamble followed by a 112-bit message.

The preamble contains four fixed pulses used for packet detection and synchronization. In PPM signaling, each bit is represented by a pulse located in either the first or second half of a 1 μ s symbol interval. Unlike continuous phase or frequency modulated waveforms, ADS-B signals are short-duration burst transmissions with sparse pulse structures. Consequently, only limited steady-state carrier behavior is observable within a single packet. Accurate temporal alignment is therefore critical as small synchronization errors can significantly affect symbol-level signal representation. Additionally, variations in received signal strength, propagation delay, and carrier frequency offset introduce distortions that complicate stable physical-layer feature extraction.

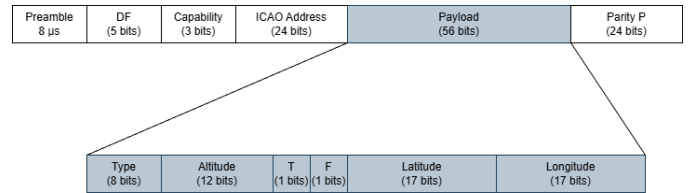


Fig. 1: Structure of a 1090ES ADS-B message.

B. Differential Constellation Traces

The differential constellation trace is a representation method that is used to reveal the temporal phase dynamics of complex baseband I/Q signals. In this approach, the relationship between successive samples of the signal is examined, and each sample is multiplied by the complex conjugate of the preceding sample. This process is mathematically defined as

$$d[n] = x[n] \cdot x^*[n-1] \quad (1)$$

where $x[n]$ denotes the complex baseband I/Q sample at discrete time index n , $x[n-1]$ is the previous complex sample, $x^*[n-1]$ represents the complex conjugate of the previous sample, and $d[n]$ is the resulting differential signal.

The complex baseband signal can be expressed as

$$x[n] = I[n] + jQ[n] = A[n]e^{j\phi[n]} \quad (2)$$

where $I[n]$ and $Q[n]$ are the in-phase and quadrature components, $A[n]$ is the instantaneous amplitude, and $\phi[n]$ is the instantaneous phase.

By substituting this expression into the differential operation, the result becomes

$$d[n] = A[n]A[n-1]e^{j(\phi[n]-\phi[n-1])} \quad (3)$$

Thus, the phase of $d[n]$ corresponds to the phase difference $\phi[n] - \phi[n - 1]$. The resulting differential signal therefore contains the phase difference between successive samples rather than absolute phase information. Consequently, this representation emphasizes phase transitions and reduces dependence on the carrier phase.

This transformation allows for a more distinct observation of temporally accumulated distortions, particularly hardware-induced phase noise, carrier frequency offset, and oscillator instabilities. Using phase difference instead of absolute phase creates a feature space that is more robust against constant phase offsets. Thus, the representation better reflects the subtle characteristics specific to the transmission chain.

In the final stage, the differential complex trace is separated into its real and imaginary components and organized into a two-channel structure. This format produces an input tensor structured along the temporal axis for convolutional architectures and enables processing through learnable filters.

Consequently, the differential constellation trace offers a more distinctive phase-based representation compared to raw I/Q samples and stands out as an effective feature extraction method, particularly in RF device recognition, emitter identification, and modulation analysis studies.

C. Convolutional Autoencoder

Convolutional autoencoders consist of two main components: an encoder and a decoder. The encoder part processes the input data through successive convolutional layers, transforming it into low, medium, and high-level feature maps. In this process, pooling layers applied after the convolutional layers reduce spatial dimensionality, lowering the number of parameters and computational cost while contributing to more distinctive representations. Activation functions provide nonlinearity to the model, enabling the learning of complex patterns. The decoder part aims to reconstruct the data using the feature maps obtained by the encoder. In this stage, transposed convolution (deconvolution) and upsampling layers are generally used to gradually increase the spatial dimensions of the data and attempt to recover the original dimensions of the input data.

Convolutional autoencoder models are trained to minimize the reconstruction error between the input and the reconstructed output. This method utilizes convolutional and pooling layers in the encoding phase, while using convolutional layers along with upsampling techniques in the decoding phase. This architecture, which can automatically extract spatial features from input data thanks to learnable filters, does not require labeled data.

IV. THREAT MODEL

ADS-B was designed as an open broadcast protocol without cryptographic authentication or encryption, making it inherently vulnerable to adversarial attacks. The widespread availability of low-cost Software-Defined Radios has significantly lowered the technical barrier to mounting such attacks, as an adversary requires only a commodity SDR platform and

basic signal processing capability to transmit fraudulent ADS-B messages [8], [11].

We consider an external adversary capable of transmitting ADS-B signals on the 1090 MHz band using hardware that differs from legitimate aircraft transponders. Under this model, the attacker may inject fabricated messages to introduce ghost aircraft into air traffic control displays, or retransmit previously recorded legitimate signals to impersonate a real aircraft. In these cases, the transmitted frames remain protocol-compliant, and therefore format-level checks and behavior-based detectors may be less reliable.

A critical observation shared across these attack classes is that the adversary's transmitting hardware is distinct from the legitimate aircraft transponder. Even when message content is faithfully reproduced, the analog signal emitted by the attacker's device carries hardware-specific imperfections including carrier frequency offset, I/Q imbalance, and oscillator instability that differ from those of the authentic source. These physical-layer discrepancies persist regardless of the transmitted data content, rendering the authentic signal difficult to replicate without access to the original RF chain. The proposed framework leverages this property to identify anomalous transmissions at the physical layer, independently of message content.

V. METHODOLOGY

A. Data Acquisition

In this study, 160 different ICAOs were selected within the dataset, where aircraft with the same ICAO address performed flights on at least three different days. This selection was made to ensure the temporal consistency of the signals from the devices and the observability of hardware-specific characteristics on different days. During the acquisition of ADS-B signals, the center frequency of the receiver device was set to 1090 MHz, and the sampling frequency was configured as 40 MHz [7].

The benign dataset consists of 71,399 samples collected from 160 aircraft across three recording sessions, of which 59,499 were used for training and 11,900 were reserved for testing. Furthermore, 59,499 malicious samples synthesized from the benign training data were appended to the test set for comprehensive evaluation.

B. Data Preprocessing

In order to ensure temporal consistency of ADS-B signals, the message start position was first determined. For this purpose, an ideal ADS-B preamble template corresponding to approximately 8 μ s at a sampling rate of 40 MHz was constructed. The template was defined such that unit amplitude values were assigned to the sample indices corresponding to the theoretical positions of the P1, P2, P3, and P4 pulses. Cross-correlation was then applied between the initial portion of the received signal and the ideal preamble template. The sample index at which the correlation output reached its maximum was identified as the message start time (T_0). This procedure enabled temporal alignment of ADS-B packets

recorded at different times by referencing a common starting point.

Using the detected T_0 as a reference, the signal segment between $T_0 + 640$ and $T_0 + 1600$ samples was extracted. At a sampling rate of 40 MHz, this interval corresponds to approximately 16–40 μ s. As a result, a fixed-length complex segment of 960 samples was obtained for each ADS-B packet. The use of fixed-length segments ensures consistent input dimensions for the learning model and prevents variability caused by unequal packet lengths.

The differential constellation trace representation was then computed over the extracted complex segments. In this step, each sample was multiplied by the complex conjugate of its preceding sample.

This transformation removes absolute phase information and instead captures the phase difference between successive samples, thereby reducing dependence on carrier phase and emphasizing temporal phase dynamics of the signal.

Subsequently, Root Mean Square (RMS)-based normalization was applied to the differential signal. For each segment, the average energy was calculated and the differential trace was scaled by the corresponding RMS value. This step mitigates the influence of amplitude scaling and makes the representation power-invariant. To avoid numerical instability, unit scaling was applied in cases where the RMS value approached zero.

Finally, the complex differential trace was separated into its real and imaginary components and organized into a two-channel structure. The resulting data representation has a dimensionality of $(N, 959, 2)$, forming an input tensor suitable for convolutional architectures.

C. Attack Data Generator

The attack dataset was generated by simulating a replay-based forgery scenario at the physical layer. In this context, the characteristics of each benign ADS-B signal were modified to obtain an attack dataset based on the assumption that previously recorded legitimate ADS-B signal segments were retransmitted by an SDR-based attacker. Furthermore, to ensure the preprocessing conditions remained identical, the same preamble detection and fixed-length segmentation steps used in the benign dataset were applied before generating the attack data.

After extracting the synchronized 960-sample complex baseband segment $x[n]$, three impairments were introduced to emulate a different transmitting device.

First, a device-specific CFO is introduced into the ideal digital baseband signal to model realistic oscillator imperfections found in practical RF hardware. Instead of generating perfectly ideal waveforms, the framework deterministically maps the target aircraft identifier to a unique seed using a hash function, ensuring that each device is assigned a stable and repeatable static frequency deviation reflecting crystal manufacturing tolerances. To better approximate real transmitter behavior, a small random variation is added during each transmission to

capture short-term instabilities. In continuous time, this effect can be expressed as

$$x_{\text{CFO}}(t) = x(t)e^{j2\pi\Delta f t}, \quad (4)$$

where Δf denotes the frequency offset in Hz. Using the discrete-time relation $t = n/f_s$, the model becomes

$$x_{\text{CFO}}[n] = x[n]e^{j2\pi\Delta f \frac{n}{f_s}}, \quad (5)$$

where f_s is the sampling frequency. This operation preserves the signal amplitude while introducing a linearly increasing phase term, corresponding to a constant angular rotation in the complex I/Q plane, thereby realistically mimicking oscillator-induced frequency drift observed in practical RF devices.

Second, I/Q imbalance was introduced to model gain and phase mismatch between the in-phase and quadrature branches. In practical RF transceivers, analog components such as mixers, amplifiers, and filters cannot maintain perfectly matched amplitude and an exact 90° phase difference between the I and Q paths due to manufacturing tolerances and hardware limitations. These imperfections introduce subtle but consistent distortions in the signal structure, making I/Q imbalance an important impairment for realistically modeling device-dependent RF characteristics. Let $I[n] = \Re\{x_{\text{CFO}}[n]\}$ and $Q[n] = \Im\{x_{\text{CFO}}[n]\}$. The impaired quadrature component was computed as

$$Q'[n] = g(Q[n] \cos \phi - I[n] \sin \phi) \quad (6)$$

where g represents gain imbalance and ϕ denotes phase error. The resulting signal becomes

$$x_{\text{IQ}}[n] = I[n] + jQ'[n]. \quad (7)$$

Finally, additional complex white Gaussian noise $w[n]$ was added to represent the unavoidable background noise present in real RF systems. In practical scenarios, received signals are always affected by random disturbances from electronic components and the environment. The noise variance was selected proportional to the average signal energy of the segment. This can be expressed as:

$$x_{\text{attack}}[n] = x_{\text{IQ}}[n] + w[n] \quad (8)$$

The resulting attack signals preserve the original ADS-B packet content while reflecting transmitter-related physical-layer deviations. These segments were then processed using the same differential constellation trace extraction and RMS normalization steps.

D. Classification

To select the most appropriate detection model, three unsupervised approaches were evaluated on the same dataset: a convolutional autoencoder (CAE), Local Outlier Factor (LOF), and One-Class Support Vector Machine (OCSVM). All models were trained exclusively on benign ADS-B signal representations.

The input to CAE network consists of differential constellation trace representations with dimensions $(N, 959, 2)$. Before training, the minimum and maximum values were computed over the benign dataset. All samples were then normalized using min-max scaling:

$$X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad (9)$$

The model follows a 1D convolutional encoder-decoder structure. The encoder contains two convolutional layers with 32 and 16 filters (kernel size 3, ReLU activation), each followed by max-pooling with stride 2 to reduce the temporal dimension. A bottleneck layer with 8 convolutional filters provides a reduced feature representation. The decoder reconstructs the signal using convolutional layers followed by upsampling operations to restore temporal resolution. The final layer applies linear activation to produce a two-channel output with the same structure as the input. When necessary, a cropping operation is used to ensure that the reconstructed signal matches the original input length.

The model is trained using the Adam optimizer with mean squared error (MSE) as the loss function. For an input sample $X \in \mathbb{R}^{L \times C}$ and its reconstruction $\hat{X}$, the reconstruction error is defined as

$$\text{MSE}(X, \hat{X}) = \frac{1}{LC} \sum_{n=1}^L \sum_{c=1}^C \left(X(n, c) - \hat{X}(n, c) \right)^2 \quad (10)$$

where L denotes the temporal length and C is the number of channels. During inference, this reconstruction error is computed for each sample. Samples with higher reconstruction error are considered anomalous.

E. Evaluation Method

Each model is evaluated on a binary classification task, where benign samples form the negative class and replay-based attack samples form the positive class. For each detector, an anomaly score is produced for every sample (CAE reconstruction error, LOF novelty score, or OCSVM decision score). A decision threshold is chosen using only benign validation data.

Performance was evaluated using accuracy, precision, recall, and F1-score. Performance metrics were calculated from the confusion matrix. This matrix illustrates the number of attack samples correctly detected (TP), benign samples correctly classified (TN), benign samples incorrectly labeled as attacks (FP), and attack samples that were not detected (FN).

Accuracy shows the overall proportion of correctly classified samples:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (11)$$

Precision indicates how many of the samples labeled as attacks are actually attack samples:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (12)$$

Recall gives the ratio of detected attack samples to the total number of actual attack samples:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (13)$$

The F1-score combines precision and recall into a single value:

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (14)$$

VI. EXPERIMENTAL RESULTS AND DISCUSSION

Table I presents the performance comparison of the evaluated unsupervised models. Overall, the CAE consistently outperforms LOF and OCSVM under the same training setup and evaluation protocol.

The CAE achieves the best overall performance, with an accuracy of 0.962 and an F1-score of 0.976. Notably, it attains near perfect precision with a high recall of 0.956, indicating that, under the selected decision threshold, no benign samples were flagged as attacks. This operating point suggests a favorable tradeoff between false alarms and missed detections for the considered dataset and attack model.

In contrast, LOF exhibits a different behavior. Although its precision remains high (0.990), the recall drops significantly to 0.188. This implies that while the model rarely misclassifies normal samples as anomalies, it fails to detect a large portion of true positive cases, resulting in a lower F1-score (0.316) and reduced overall accuracy (0.322). Such behavior indicates an overly conservative decision boundary.

Similarly, OCSVM demonstrates high precision (0.964) but suffers from limited recall (0.272), leading to moderate performance with an F1-score of 0.424 and an accuracy of 0.385. Compared to LOF, OCSVM detects slightly more positive samples; however, it still struggles to capture the majority of true anomalies, as reflected by the high false negative rate.

The superior performance of the Convolutional Autoencoder can be attributed to the structural properties of the input representation, namely the differential constellation traces. These traces inherently contain structured patterns arising from oscillator mismatch, phase noise, and I/Q imbalance, which emerge as consistent yet subtle distortions in the phase evolution of the signal.

Given these structural characteristics, convolutional architectures are particularly well-suited for such data, as they are designed to capture local dependencies and translation-invariant features. By applying learnable convolutional filters across the differential traces, the autoencoder can effectively model the underlying phase continuity and device-specific micro-structures embedded in the signal. In contrast, classical unsupervised methods such as LOF and OCSVM rely primarily on global distance metrics or boundary-based separation in feature space, which limits their ability to capture high-dimensional nonlinear dependencies present in structured RF data.

Overall, these results indicate that the CAE provides a more balanced operating point with high precision and high recall,

supporting the suitability of DCT-driven convolutional reconstruction for unsupervised physical-layer intrusion detection in the evaluated ADS-B setting.

VII. CONCLUSION

This study presented an unsupervised physical-layer intrusion detection framework for ADS-B systems, leveraging a convolutional autoencoder architecture applied to Differential Constellation Trace (DCT) features. In contrast to cryptographic enhancements or infrastructure-dependent solutions, the proposed Radio Frequency Fingerprinting-based intrusion detection framework requires neither ADS-B protocol modifications nor alterations to the existing infrastructure. Furthermore, its unsupervised nature eliminates the reliance on labeled attack data, making it exceptionally well-suited for operational aviation environments where stringent regulatory constraints severely restrict active vulnerability testing.

By leveraging differential processing, the framework captures transmitter-specific phase dynamics while reducing sensitivity to constant phase offsets, amplitude scaling, and receiver-dependent distortions. This representation enables the extraction of stable hardware-induced characteristics even under the inherently discontinuous transmission pattern of ADS-B signals.

The capabilities of the proposed framework were validated using a real-world passive dataset collected from 160 civil aircraft. To evaluate the system's robustness, a replay-based attack scenario was constructed by retransmitting legitimate ADS-B packets from this dataset through a simulated adversarial model. While the message content remained protocol-compliant and unchanged, the physical-layer characteristics were deliberately altered by introducing realistic carrier frequency offset variations, I/Q imbalance, and additive noise. This process effectively modeled the hardware differences between a legitimate aircraft transponder and an attacker's transmitter. Under these realistic attack conditions, the proposed method achieved a high detection performance, yielding an accuracy of 96.2% and an F1-score of 97.6%.

Furthermore, comparative analysis with classical unsupervised methods demonstrated that the convolutional autoencoder achieves a significantly superior balance between precision and recall, indicating its specific effectiveness in modeling the subtle radio frequency fingerprint patterns embedded within DCT features. These results confirm that the proposed method can reliably identify anomalous transmitter-level differences without requiring protocol modifications or prior knowledge of attack patterns, even when malicious transmissions strictly adhere to valid protocol rules and realistic flight behaviors. Consequently, these findings establish that reconstruction-based deep architectures are exceptionally well-suited for structured radio frequency representations, providing a robust and practical approach to fortifying ADS-B security at the physical layer.

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TABLE I. Performance Comparison of Unsupervised Methods

Method	TN	FP	TP	FN	Accuracy	Precision	Recall	F1-Score
Convolutional Autoencoder	11796	104	56899	2600	0.962	0.998	0.956	0.976
LOF	11789	111	11189	48310	0.322	0.990	0.188	0.316
OCSVM	11287	613	16165	43334	0.385	0.964	0.272	0.424